

SCORING, SUBSTITUTION MATRICES AND SCORE AGGREGATION FOR TCR ANTIGEN-SPECIFICITY SEARCH

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vdjmatch technical appendix

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (CASS...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

1. THE SEARCH-AND-SCORE FRAMEWORK

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \text{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR ($=/X/I/D$), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§3).

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2. SUBSTITUTION SCORING AND THE VDJAM MATRIX

How BLOSUM IS BUILT, AND WHY IT DOES NOT TRANSFER.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab} ; (iv) score $s(a, b) = \frac{1}{\lambda} \log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a . The signal is “which residues substitute for one another within conserved, column-homologous positions,” and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so “alignment columns” do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins CASS and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal CASS is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

WHAT VDJAM CHANGES (POINT BY POINT).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only “aligned residue pair” obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a , a *position-specific* background $\text{bg}(\cdot | L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM’s percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are “expected” and contribute little), and prospectively by the control-calibrated significance layer (§3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

OBSERVATION UNIT AND ESTIMATOR.. We replace “alignment columns” with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c , let $\text{bg}(\cdot | L, p)$ be the residue frequency at position p among all length- L chain- c CDR3s. With $f(a, b)$ the observed count of $\{a, b\}$ events and

$$e(a, b) = \sum_{\text{event slots } (L, p)} n_{L, p} [\text{bg}(a | L, p) \text{bg}(b | L, p) + \text{bg}(b | L, p) \text{bg}(a | L, p)], \quad \text{VDJAM}(a, b) = \log_2 \frac{f(a, b) + \alpha}{e(a, b) + \alpha}, \quad (1)$$

where $n_{L,p}$ is the number of observed events at (L,p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and “expected” (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\text{pen}(a,b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

REGION STRUCTURE: NDN IS LEARNED, V/J ARE GERMLINE-DETERMINED.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank “substitutions” are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \text{Pr}[\text{position } p \text{ is germline-retained}]$, determined by the gene’s trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via `mirpy’s mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the CASS V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by $1 - \text{Pr}[\text{germline}]$ and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

3. SCORE AGGREGATION AND SIGNIFICANCE

PER-HIT TO PER-QUERY.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

CONTROL-CALIBRATED E-VALUE (PRIMARY).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query’s VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M) n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

PAIRED $\alpha\beta$ E-VALUE.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural

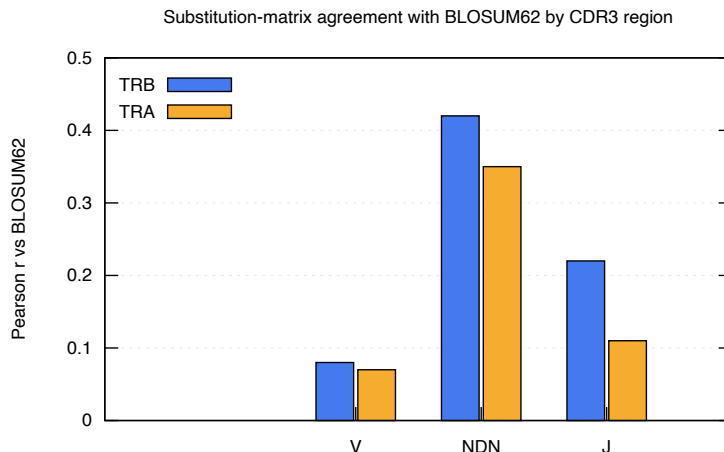


Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

and paired-sequencing data [4], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^\alpha \pi_0^\beta$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N \pi_0^\alpha \pi_0^\beta$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain’s, true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9} . (When the two chains’ hits are treated as independent tests, the equivalent statistic is Fisher’s combination of the per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

4. RESULTS: DOES VDjam (AND REGION WEIGHTING) HELP?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDjam, and region-weighted NDN-VDjam — ranking by score and reporting micro PR-AUC for “hit shares E^* ” (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* (+0.004 at scope 2, +0.007 at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution*

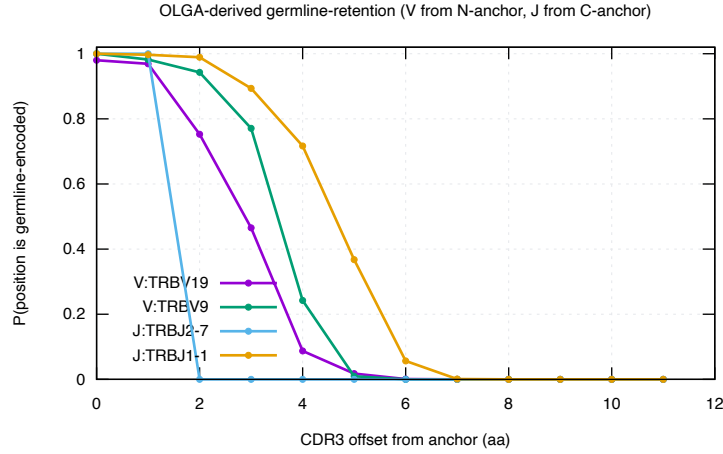


Figure 2: OLGA-derived germline-retention $w(p)$ (via *mirpy*): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

matrix at this narrow-scope retrieval task (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§3).

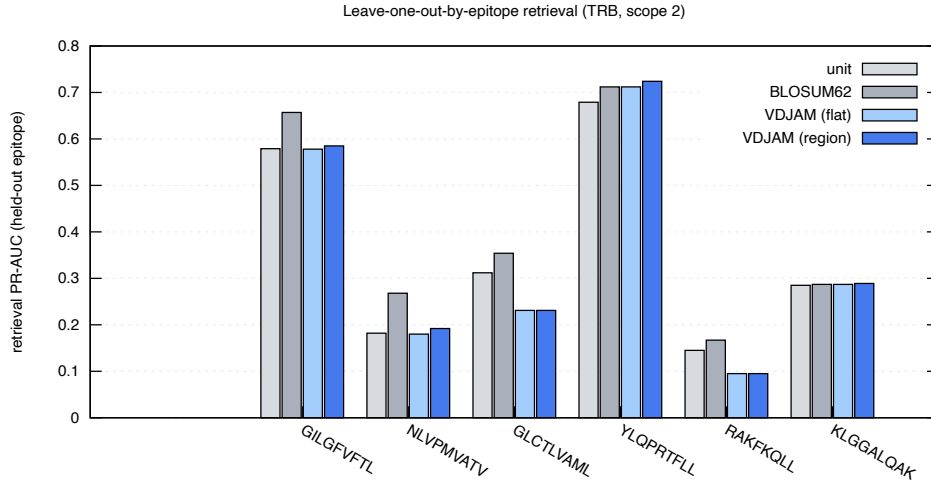


Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

5. EDIT DISTANCE, POSITION, AND THE MATRIX COMPARISON

Four analyses (`bench/scoring_analysis.py`, `bench/loo_vdjam.py`) settle what scoring actually buys on VDJdb. All figures and numbers below are the *human TRB* example case (the largest, best-populated slice of VDJdb); the same pipeline applies unchanged to human/mouse and to TRA/TRB — point `$VDJDB_SAMPLE/$VDJDB_SPECIES` at another export to regenerate.

HAMMING DISTANCE 1 IS THE SIGNAL:NOISE OPTIMUM.. The fraction of CDR3 neighbours that share an epitope — the precision of a similarity call — falls steeply with edit distance: macro purity (per-epitope mean, robust to the $100\times$ epitope-size range; § below) 0.49, 0.24, 0.15, 0.10, 0.07 at distance 1 ... 5, an enrichment of $56\times \rightarrow 2.5\times$ over chance (Fig. 5). A distance-1 neighbour is twice as likely to be a true co-specific as a distance-2 one and seven times a distance-5 one. This reproduces the original VDJdb observation [6] that single-mismatch neighbours are the sweet spot — and on the dense 2026 release it survives only once a few 10x mega-studies are capped and the random tail is treated as an admixed control (§3); the naive pooled rate reads a near-constant ~ 0.9 because a random clonotype is probably from a 30k-clonotype epitope. It frames everything below: the search ball, not the substitution matrix, is the dominant lever; the matrix only re-ranks within a ball whose purity is already set by its radius.

CENTRAL SUBSTITUTIONS CARRY THE SPECIFICITY SIGNAL.. Conditioning on *where* a single substitution falls (Fig. 6), the probability that the two CDR3s still share an epitope dips to ≈ 0.31 in the NDN core but rises to ≈ 0.75 within a few residues of either the V or J anchor: a central mismatch most often *changes* specificity (it is contact-proximal and informative), while a near-anchor mismatch is usually germline noise (the conserved V/J stems). This is the empirical basis for weighting the centre more — the same direction as the germline-retention weight $1 - w(p)$ (§2) — and, encoded directly as a positional weight on BLOSUM62 (the PSSM of §5), it is the one change that clearly improves retrieval over flat BLOSUM62 (7/8 epitopes; Table 1). It also matches the contacting-region picture of Shcherbinin et al. [5].

THE NDN CORE IS GLYCINE-RICH.. Central CDR3 positions reach $\sim 31\text{--}34\%$ glycine in both chains (Fig. 7) — the N-region/D-gene insertion signature — a strongly non-BLOSUM composition. Notably TRA is comparably central-G-rich to TRB, so this is an insertion-diversity feature, not a TRB-D-gene-exclusive one; either way the NDN alphabet is distinctive enough that a general matrix is a poor fit there in principle.

NO AMINO-ACID MATRIX CLEARLY BEATS BLOSUM62 — AND A GENETIC-CODE NULL TIES IT.. Re-scoring the held-out-epitope candidates (leave-one-out, § above) with each substitution matrix, scored by balanced PR-AUC on the composition-controlled long list, gives a consistent verdict at every scope (Table 1, Fig. 4): BLOSUM62, PAM250 and the structural matrix are essentially tied at the top, plain Hamming (unit) is close behind, and the data-derived NDN-VDJAM — flat or region-weighted — trails them all. Strikingly, **VDJAMr**, a matrix built from the *genetic code alone* (`loo_vdjam.codon_dissim`: $M(a, b) = \Pr[\text{a single random nucleotide substitution of a random codon of } a \text{ yields } b]$, with no chemistry or selection), *matches* BLOSUM62 (+0.017 at distance ≤ 2 , +0.018 at ≤ 4 ; ahead in 5–6/8 epitopes) and clearly beats the data-derived VDJAM. That a pure codon-mutational-accessibility null does as well as a chemically tuned matrix says the substitution structure of co-specific CDR3s

is largely *generative* (what the recombination/mutation process reaches cheaply), not chemical — which is exactly why hand-tuning the amino-acid alphabet does not help. The substitution matrix is a second-order lever; the first-order statistic remains the control-calibrated neighbour count at distance 1 (§3).

BUT POSITION-WEIGHTING BLOSUM62 DOES (THE SIGNIFICANCE PSSM).. Experiment 2 says *where* a substitution falls matters more than *which* substitution it is. We encode that as a positional factor on top of BLOSUM62. Crucially the profile is **anchored from both ends**, not by relative position: the germline V/J stems sit at fixed *absolute* offsets from the CDR3 termini regardless of length, so we learn two end-anchored profiles $\text{Pr}(\text{same} \mid \text{sub at offset } d \text{ from the V anchor})$ and the J-anchor counterpart (Fig. 6), Beta-Binomial-smoothed so sparse deep-core offsets shrink toward the global rate (`regions.load_significance`; estimated by `bench/scoring_analysis.py`). The per-position weight is $\omega(p) = 1 - \max(\text{Pr}_V(d_V), \text{Pr}_J(d_J))$ with $d_V = p$, $d_J = L - 1 - p$, normalised to mean 1: a position is down-weighted if it lies near *either* anchor, up-weighted in the NDN core. This is the data-learned twin of the germline-retention weight (both $1 - \max(\text{V-side}, \text{J-side})$). `seqtree` consumes it natively — `regions.significance_pssm()` builds `PositionalMatrix.from_weights(BLOSUM62, [100  $\omega(p)$ ])`, so $\text{pen}(p, a, b) = \omega(p) \cdot \text{BLOSUM62}(a, b)$. The multiplicative form is justified by the data: stratifying single substitutions by anchor offset $\times$ BLOSUM severity, substitution severity matters *only in the core* ($\text{Pr}(\text{same})$ 0.50 conservative vs 0.38 radical) and is flat near the anchors — BLOSUM discriminates exactly where ω is large. This is the only scoring change that clearly beats flat BLOSUM62, winning 7/8 held-out epitopes at distance ≤ 2 and 6/8 at ≤ 4 (Table 1); the gain is uniformly signed because the matrix re-ranks within a ball whose purity is already set by the radius. The remaining lever, the *V-gene* dimension, is taken up in §6.

A PURPOSE-BUILT TCR MATRIX (TCRBLOSUM) DOES NOT TRANSFER.. Postovskaya et al. [3] report that `terBLOSUM` — a BLOSUM-style matrix counted from same-epitope, equal-length CDR3 pairs — improves epitope-specific TCR clustering over BLOSUM62. Two features make that gain suspect: the counts apply *no* redundancy clustering (so convergent public clonotypes dominate the statistics), and the matrix is validated on the same same-epitope relation it is fit to. Run through our leave-one-out-by-epitope retrieval (their published `terBLOSUMb`, `bench/tcrblosum_refute.py`), it does *not* beat BLOSUM62: 0.555 vs 0.567 at distance ≤ 2 (ahead in 2/8 epitopes) and essentially tied at ≤ 4 (0.589 vs 0.591), while their bundled BLOSUM62 reproduces ours exactly (a loader sanity check). Reweighting *either* matrix by the central-significance PSSM helps, and it helps BLOSUM62 the most (BLOSUM+possig 0.603 vs `terBLOSUM`+possig 0.598). The lesson is consistent: a re-counted amino-acid alphabet does not transfer across epitopes; position does.

6. THE V-GENE DIMENSION

The remaining lever is the V gene. Three questions on the composition-controlled long list (human TRB unless noted): is a V match itself a specificity signal, does it soften to germline-similar V genes, and — if so — does it localise to a few contacting-loop positions (a V “pseudosequence”)?

A V MATCH IS A STRONG, NEAR-BINARY SPECIFICITY PRIOR.. Splitting every scored neighbour pair by whether query and hit use the same V family, same-V neighbours share the held-out epitope 44%

balanced PR-AUC	unit	BLOSUM62	PAM250	structural	VDJAM	VDJAMr	VDJAM-region	BLOSUM+possig
edit distance ≤ 2	0.543	0.564	0.572	0.564	0.524	0.581	0.529	0.598
edit distance ≤ 4	0.573	0.585	0.590	0.585	0.535	0.603	0.543	0.611

Table 1: Substitution matrices by leave-one-out-by-epitope VDJdb-vs-VDJdb retrieval (human TRB, 8 largest epitopes of the 2026-06-11-ZENODO composition-controlled long list; balanced PR-AUC). The chemical matrices (BLOSUM62/PAM250/structural) and the genetic-code null VDJAMr are all $\approx$ tied; the data-derived VDJAM trails; only BLOSUM62 reweighted by the positional-significance PSSM clearly leads (7/8 and 6/8 epitopes).

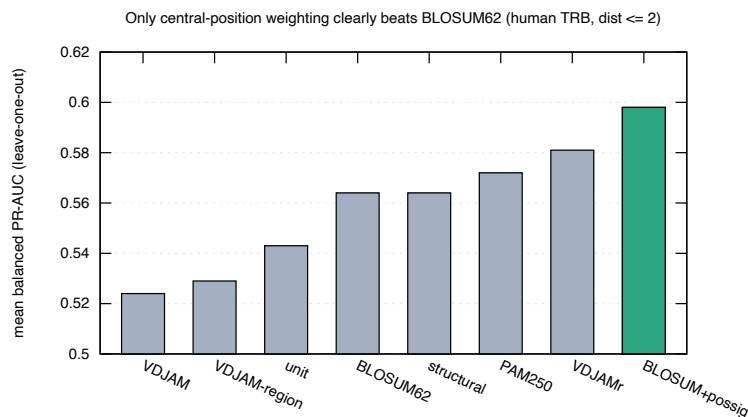


Figure 4: Mean leave-one-out retrieval PR-AUC by scoring (TRB, distance ≤ 2). No standard amino-acid matrix beats BLOSUM62; reweighting BLOSUM62 by the central-position significance factor (green) does.

of the time versus 6% for cross-V (edit distance ≤ 4 ; Table 2) — a $\sim 7\times$ prior, far larger than any substitution-matrix effect. This is why the legacy tool used V as a hard filter, and it argues for carrying V as a strong prior (a soft $S(V)$ term or a per-V null) in a joint V+CDR3 score. Position weighting sharpens the *same-V* stratum most (balanced PR 0.632 $\rightarrow$ 0.713 with the significance PSSM, +0.081) — where the germline frame is shared and only the NDN core varies — and little in the noise-dominated cross-V stratum (+0.012).

COARSE GERMLINE-LOOP SIMILARITY BARELY PREDICTS IT.. If the same-V advantage were simply the chemistry of the V-encoded contacts, cross-V co-specificity should rise smoothly with germline CDR1+CDR2 similarity. As a *continuous* quantity it barely does: across every {human,mouse} $\times$ {TRA,TRB} $\times$ {MH I,MHC-II} cell (bench/vgene_scan.py, Fig. 8, Table 4) the V prior is universal (same/cross ratio 1.0–8 $\times$) but the point-biserial correlation between loop similarity and co-specificity is ≈ 0 ($r \in [-0.06, +0.06]$ in the well-powered cells). Binning cross-V pairs by similarity shows only a weak rise — 6% at low similarity to 12% at the top bin, still far below the same-V 44% (Table 3). So a loose, continuous “fuzzy-V” similarity does not recover the prior.

NEAR-EXACT GERMLINE IDENTITY PARTLY RECOVERS IT — AT CDR3-LIKE TOLERANCE.. Resolving the similarity to exact germline-region matches changes the picture (bench/vregion_decompose.py, hu-

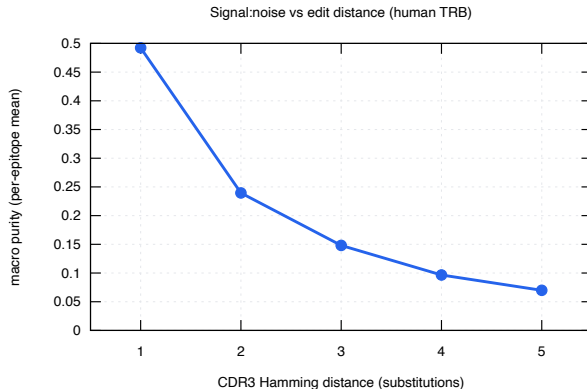


Figure 5: Macro purity (per-epitope mean fraction of CDR3 neighbours sharing an epitope) vs Hamming distance (human TRB, composition-controlled long list): distance 1 is the signal:noise optimum [6].

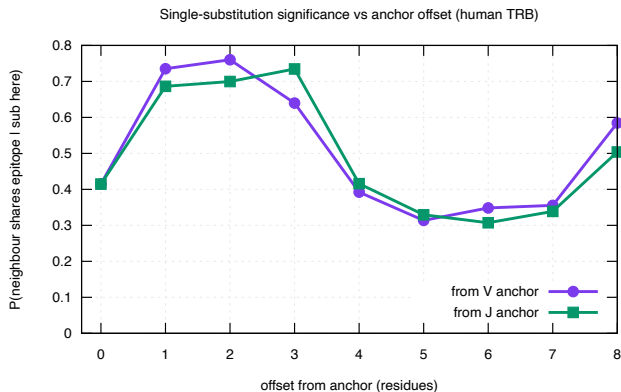


Figure 6: $\Pr(\text{neighbour shares epitope} \mid \text{single substitution at relative position})$, human TRB. The NDN core is where a mismatch most changes specificity; the V/J borders are germline noise.

man TRB MHC-I; Table 5). Cross-V co-specificity rises *monotonically* as the germline CDR1+CDR2 edit distance shrinks — 10.5% at ≥ 6 mismatches, 12.6% at 3–5, 17.9% at 1–2, and 32% at 0 (identical loops): a real $\sim 3\times$ lift over the cross-V floor. FR1 identity behaves likewise (27.6%, $2.6\times$). So the germline contacting loops *do* carry part of the V signal, at a tight CDR3-like tolerance (near-exact identity; the continuous metric above, dominated by 1–2-mismatch pairs, misses it) — but even identical loops recover only about *half* the same-V advantage (32% vs 64%); the rest is gene-identity-specific (the exact CDR3-germline anchor and integrated docking geometry). Caveat: the identical-loop bin is small ($n = 25$ at distance ≤ 2 , $n = 5$ at ≤ 1 , and bounced 32–60% across held-out sets) — the robust claim is the monotone gradient over the well-powered bins ($n = 507, 3735, 36853$), not the exact endpoint.

THE RECOVERY IS WHOLE-LOOP, NOT A SPARSE PSEUDOSEQUENCE.. Do a few decisive positions carry it (the MHC-pseudosequence analogue)? Per-position residue-match lift over the cross-V baseline

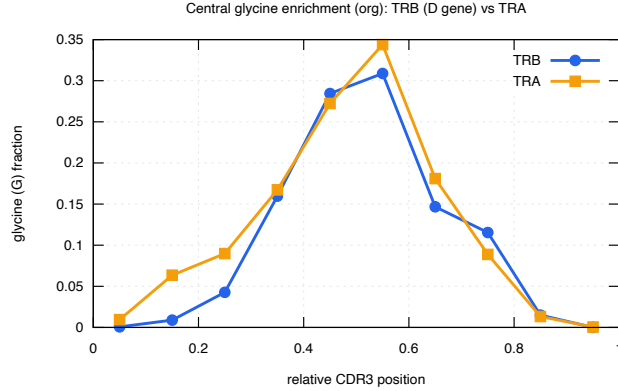


Figure 7: Glycine fraction by relative CDR3 position (human): the NDN core is $\sim 30\text{--}36\%$ glycine in both TRA and TRB (insertion/D-gene signature).

(`bench/vpseudo.py`, length-constant CDR1/CDR2) is mostly flat (lift 1.0–1.2 at the 10.5% baseline), with a single exception — CDR2 position 5 ($1.56\times$, $n = 4833$). So the partial recovery above is *whole-loop* near-identity, not the matching of a handful of pseudosequence coordinates: a different V gene must reproduce essentially the entire CDR1+CDR2 to behave like the same V. A useful soft-V match must therefore demand near-exact loop identity, not loose similarity. The next step is the full IMGT-numbered treatment (FR positions and the FR3 DE-loop/“CDR2.5”, via the *arda* IMGT/V-QUEST milestone) to test whether specific positions sharpen this.

V stratum	pairs	co-specific	flat	+possig	+region
same V	120 738	45.2%	0.626	0.682	0.678
cross V	979 199	6.2%	0.558	0.569	0.538

Table 2: V-gene-stratified leave-one-out retrieval (human TRB, edit distance ≤ 4 ; co-specific = positive rate, the rest balanced PR-AUC). A V match is a $\sim 7\times$ co-specificity prior; the significance PSSM helps the same-V stratum most (+0.056).

germline CDR1+CDR2 similarity	cross-V pairs	co-specific
[0.0, 0.3)	479 181	5.8%
[0.3, 0.5)	387 839	6.2%
[0.5, 0.7)	89 533	7.4%
[0.7, 0.9)	18 320	10.2%
[0.9, 1.0)	4 326	11.6%

Table 3: Continuous germline CDR1+CDR2 similarity gives only a weak rise in cross-V co-specificity (6% $\rightarrow$ 12%), far below the same-V 45% — which is why $r \approx 0$. The stronger signal lives at near-*exact* identity (Table 5), not along the continuous similarity axis.

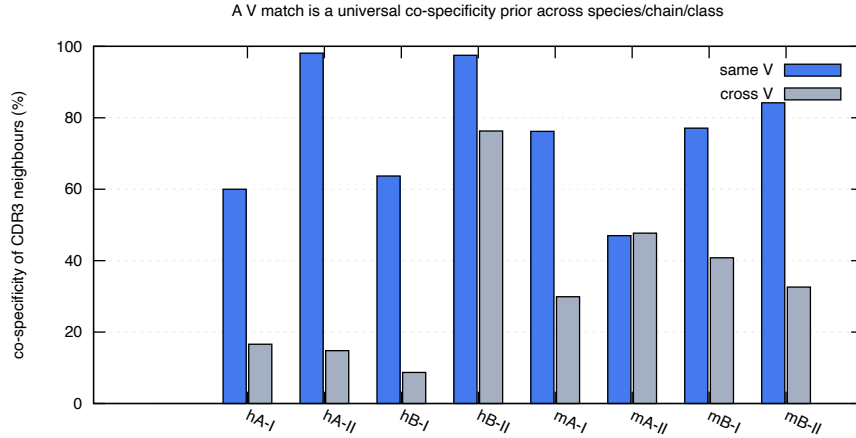


Figure 8: Same-V vs cross-V CDR3-neighbour co-specificity across species/chain/MHC-class cells of VDJdb (h/m = human/mouse; A/B = TRA/TRB; I/II = MHC class). A V match is a universal co-specificity prior; continuous CDR1/CDR2 similarity does not bridge the gap (Table 4).

cell (species / chain / class)	same-V	cross-V	ratio	$r(\text{vsim, co-spec})$
human TRA MHC-I	60.0%	16.6%	3.6	0.050
human TRA MHC-II	98.1%	14.8%	6.6	0.269
human TRB MHC-I	63.7%	8.7%	7.3	0.070
human TRB MHC-II	97.5%	76.3%	1.3	0.054
mouse TRA MHC-I	75.1%	29.3%	2.6	0.021
mouse TRA MHC-II	47.0%	47.7%	1.0	−0.009
mouse TRB MHC-I	77.1%	40.8%	1.9	−0.080
mouse TRB MHC-II	84.2%	32.6%	2.6	−0.017

Table 4: Per-cell same-V vs cross-V co-specificity and the point-biserial correlation between continuous germline CDR1+CDR2 similarity and co-specificity among cross-V pairs (top 6 epitopes/cell, ≥ 30 CDR3/epitope). The V prior holds everywhere (ratio 1–8 $\times$); continuous loop similarity barely predicts ($r \approx 0$; the 0.24 is human TRA MHC-II, whose high-similarity bin has $n = 4$).

7. BENCHMARK: THE REFERENCE-REPLICATED SHORTLIST

The latest VDJdb release is our standing benchmark, but its labels vary in confidence. The cleanest gold standard is a clonotype–epitope association reported independently in ≥ 2 references (`db.replicated`, null references excluded; ~ 3000 TRB and ~ 2000 TRA such pairs in the current human release) — the same “shortlist” idea used for `mhcmatch`. We measure annotation accuracy on it by leave-one-out nearest-neighbour voting against the rest of VDJdb at the signal:noise-optimal scope (subs 1), excluding the clonotype’s own exact CDR3 so the label must come from *other* TCRs (`bench/shortlist_accuracy.py`). Replicated TCRs are annotated at ~ 44 – 47% top-1, modestly above size-matched single-reference controls of the same epitopes (Table 6). The dramatic $3\times$ gap we saw on a sparse older export was an artefact: on the dense 2026 release even single-reference clonotypes have findable neighbours (control recall ~ 48 – 72%), so the gold set is cleaner but no longer dramatically *easier*. Restricting

cross-V stratum (human TRB MHC-I)	co-specific	<i>n</i>
<i>cross-V baseline</i>	10.8%	41 132
CDR1+CDR2 edit distance ≥ 6	10.5%	36 853
edit 3–5	12.6%	3 735
edit 1–2	17.9%	507
edit 0 (identical)	32.0%	25
FR1 identical	27.6%	257
<i>same-V reference</i>	63.8%	14 883

Table 5: Cross-V co-specificity rises monotonically as germline CDR1+CDR2 approach identity; at exact identity it reaches $\sim 32\%$ — a real $\sim 3\times$ lift over the cross-V floor, but only about *half* the same-V level (64%). Germline-loop identity carries part of the V prior at a tight, near-exact tolerance; the rest is gene-identity-specific. (edit 0/CDR1&CDR2 bin small, $n = 25$.)

neighbours to the same V family helps at the wider subs 2 scope (TRB 53.1 $\rightarrow$ 56.8%) — exactly where cross-V false neighbours appear (§6), confirming the V prior is worth its place.

chain	set	<i>n</i>	top-1 (CDR3)	top-1 (V-restr.)	recall
TRA	shortlist (≥ 2 ref)	1993	43.9%	43.8%	62.3%
TRA	control (1 ref)	1993	33.0%	29.6%	47.5%
TRB	shortlist (≥ 2 ref)	2962	47.3%	47.1%	54.2%
TRB	control (1 ref)	2962	46.7%	43.7%	50.2%

Table 6: Leave-one-out epitope-annotation accuracy on the reference-replicated shortlist vs a size-matched single-reference control (human, subs 1, exact self excluded). Top-1 = nearest-neighbour-vote epitope correct; recall = true epitope reachable within scope. On the dense 2026 release the shortlist is modestly above control, not the $3\times$ of the older sparse export.

8. WORKED EXAMPLES

PAIRWISE (DIFFERENT LENGTHS).. seqtree aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved CASS V-anchor and DTQYF J-motif with substitutions and indels confined to the central NDN core:

query (ref): CASSPRSGETQYF epitope GILGFVFTL (unit / edit-distance alignment)

vs CASSNRSGETQYF (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)

```
CASSPRSGETQYF
|||| |||||
CASSNRSGETQYF
```

vs CASSIRSGESTQYF (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)

```
CASSPRSGETQYF
|||| ||| |||
CASSIRSGESTQYF
```

```

vs CASSLFSETQYF (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
CASSPRSGETQYF
|||| | ||||
CASSLFS-ETQYF

```

```

vs CASSLGETQYF (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
CASSPRSGETQYF
|||| |||||
CASS--LGETQYF

```

```

vs CASSPWSGSQETQYF (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
CASSPRSG--ETQYF
|||| | ||||
CASSPWSGSQETQYF

```

MULTIPLE (SAME LENGTH).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 9–10): columns are shaded by conservation, dark at the germline-determined CASS prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [5] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

9. LIMITATIONS AND ROADMAP

The substitution *alphabet* is a second-order lever: the data-derived NDN-VDJAM trails BLOSUM62 at every scope, and a chemistry-free genetic-code null (VDJAMr) does just as well as BLOSUM62 — the co-specific-CDR3 substitution structure is largely generative, not chemical (§5). What moves the needle is *position*: reweighting BLOSUM62 by the central-significance PSSM clearly beats flat BLOSUM62 (7/8 epitopes), confirming that for CDR3 *where* a mismatch falls matters more than *which* residue it is. The *V-gene* dimension (§6) is a strong near-binary prior ($\sim 7\times$); it is *not* captured by loose germline-loop similarity, but cross-V co-specificity does rise at *near-exact* CDR1+CDR2 identity — recovering about *half* the same-V advantage ($\sim 32\%$ vs 64%) at a tight, CDR3-like tolerance, whole-loop rather than via a sparse pseudosequence (exact-match bin small, $n \approx 25$). The open work is to model V as a strong prior in a joint V+CDR3 E-value, to demand near-exact (not loose) loop identity for any soft-V match, and to extend the position analysis to full IMGT numbering (FR positions and the FR3 DE-loop, via the **arda** IMGT/V-QUEST milestone); a per-chain treatment (TRA vs TRB) is also pending. Separately, the control’s V–J usage must match the query repertoire’s or the null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes.

A further direction is to predict, *a priori*, which epitopes are intrinsically easy versus hard to annotate — the wide spread of per-epitope PR-AUC in our experiments (e.g. the convergent, “fea-

GILGFVFTL (TRB CDR3, length 13, n=14)

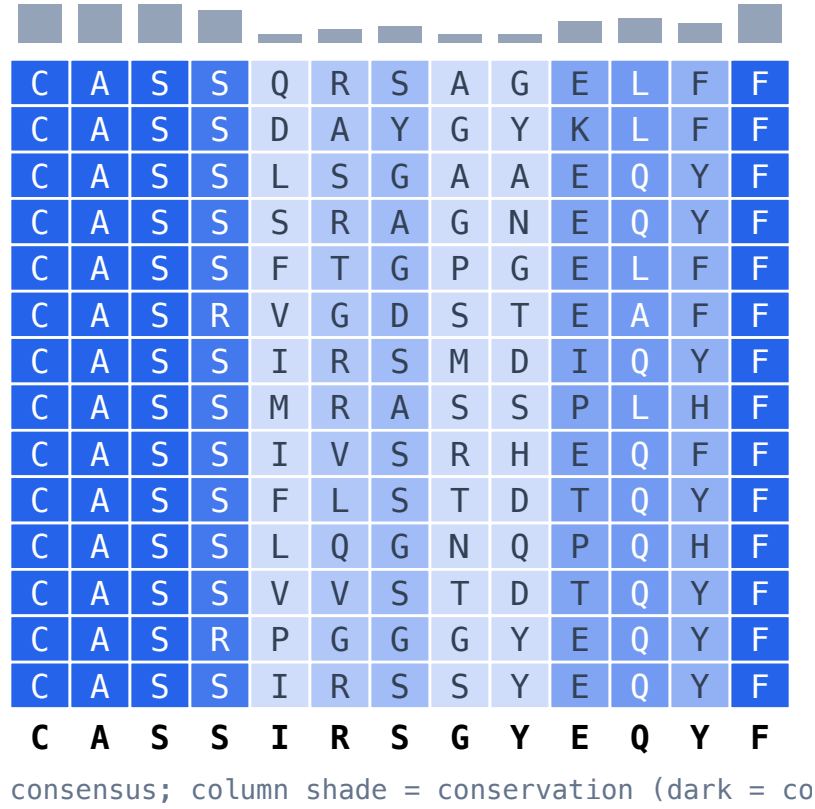


Figure 9: GIL (influenza M1, A^{*}02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

tured” CMV NLV vs the diffuse, “featureless” influenza GIL) is not noise but biology. A pMHC with prominent TCR-facing features selects a focused, motif-rich repertoire that clusters cleanly, whereas a featureless one is recognised by structurally diverse TCRs in varied docking modes and yields no tight motif [8, 9, 7, 2]; clustering-based specificity inference succeeds or fails accordingly [1]. A model that scores epitope “annotability” from repertoire convergence (and, where available, pMHC structural features) would let us report calibrated confidence per epitope and triage GIL-like hard cases — a planned addition to the benchmark. These extensions, and the full comparison against existing tools, are ongoing work.

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NLVPMVATV (TRB CDR3, length 15, n=14)

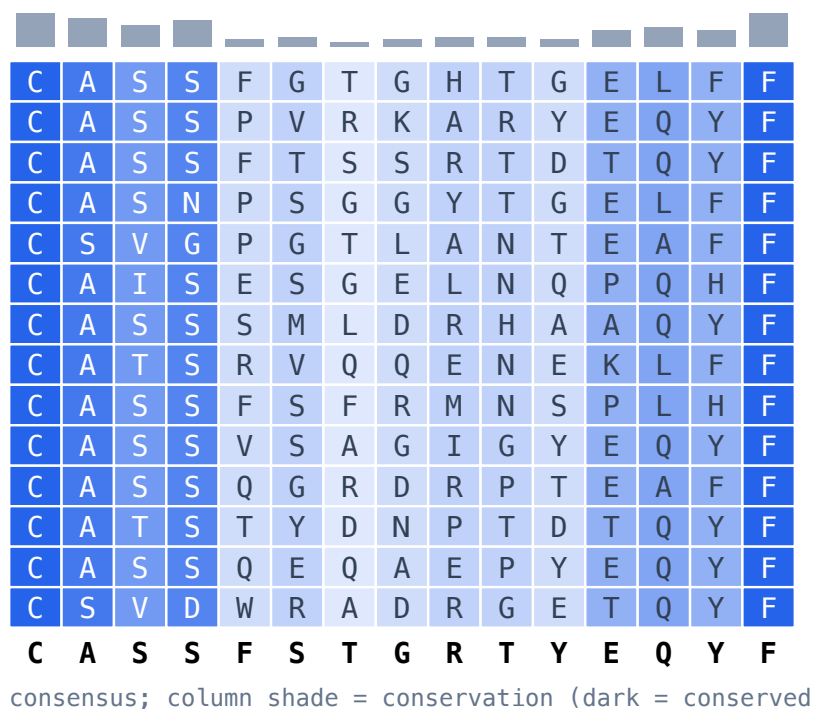


Figure 10: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

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